Asymptotic Analysis Comprehensions

Buce-Ithon

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Contents

Figure 1: From order evaluating to asymptopia ¡—

1 Basics of order, $\mathcal{O}$ and o

1.1 Summary/Comprehensions

The knowledge of orders of 2 infinitesimal or infinite quantities and the connection between them are common for us. Here we list some notes you may need to pay attention to:

1. Sometimes the "big $\mathcal{O}$ constant" in a strong function can be related to the argument, and the same is true for the small "o".

Example. $\sin(xy) = O(1)$, $\sin(xy) = O_y(x)$

2. The infinitesimal of 2 functions can't always keep for their inverse functions, i.e. $\varphi(x) = o(f(x)) \nRightarrow \widetilde{f}(x) = o(\widetilde{\varphi}(x))$. (Here: $\varphi(x)$ and $f(x)$ are increasing functions, as $\widetilde{\varphi}(x)$ and $\widetilde{f}(x)$ are their inverse functions)

Example. $f(x) = e^x, \quad \varphi(x) = \frac{e^x}{x}, \quad x \to \infty$

3. The elementary of $\mathcal{O}$ and o notation:

Notes. $\varphi = O(f)$ means an inequality, while $\varphi = o(g)$ express a limitation.

4. 16 Rules of calculating $\mathcal{O}$ and o:

1. Transitivity

   [1] $\lim_{x\to x_0} f(x) = \infty$, $\varphi(x) = O(1) \implies \varphi(x) = o(f(x))$

   [2] $f = O(\varphi)$, $\varphi = O(\psi) \implies f = O(\psi)$

   [3] $f = O(\varphi)$, $\varphi = o(\psi) \implies f = o(\psi)$

2. $\mathcal{O}$

   [4] $O(f) + O(g) = O(f+g)$

   [5] $O(f)O(g) = O(fg)$

3. $\mathcal{O}$ & o

   [6] $o(1)O(f) = o(f)$

   [7] $O(1)o(f) = o(f)$

   [8] $O(f) + o(f) = O(f)$

4. o

   [9] $o(f) + o(g) = o(|f| + |g|)$

   [10] $o(f)o(g) = o(fg)$

5. Power

   [11] $(O(f))^k = O_k(f^k),\ k \in \mathbb{N}$

   [12] $(o(f))^k = o(f^k)$

6. Equivalence $\sim$

   [13] $f \sim g$, $g \sim \varphi \implies f \sim \varphi$

6. Equivalence $\sim$

   [14] $f = o(g),\ g \sim \varphi \implies g \sim \varphi \pm kf$

7. Summation of o

   [15] $f,\ g > 0,\ f = o(g),\ x \to \infty \implies \int_A^B f(x)dx = o(\int_A^B g(x)dx)$, $B > A \to \infty$
   Specially, if $\exists A_0$ s.t. $\int_{A_0}^{\infty} g dx < \infty$, then $\int_A^{\infty} f(x)dx = o(\int_A^{\infty} g(x)dx)$, $B > A \to \infty$

   [16] $a_n,\ b_n > 0,\ a_n = o(b_n),\ n \to \infty \implies \sum_{n=N}^{M} a_n = o(\sum_{n=N}^{M} b_n)$, $M > N \to \infty$
   Specially, if $\sum_{n=1}^{\infty} b_n < \infty$, then $\sum_{n=N}^{\infty} a_n = o(\sum_{n=N}^{\infty} b_n)$, $N \to \infty$

1.2 Problem Sets

Problem. (Infinitesimal calculation) $\int_{x^{-2}}^{\infty} e^{-t}\log t\,dt = o(1),\ x \to \infty.$

> *Solution.* $\int_{x^{-2}}^{\infty} e^{-t}\log t\,dt < \int_{x^{-2}}^{\infty} e^{-\frac{t}{2}}\,dt = -2e^{-\frac{t}{2}}|_{x^{-2}}^{\infty} = 2e^{-\frac{1}{2x^2}}$

Problem.

$$\text{(Classical limit review)}x^A = o((1+a)^{\epsilon x}),\ x \to \infty,\ \text{as } \epsilon, a > 0\ \forall A.$$
$$\text{(Develop)}(\log(x))^A = o(x^{\epsilon}),\ x \to \infty,$$
$$(f(x))^A = o(e^{\epsilon f(x)}),\ x \to \infty.$$

> *Solution.* Let $n \triangleq [x],\ m \triangleq [A]+1,$ then $(1+a)^x \geq (1+a)^n \geq C_n^{m+1}a^{m+1}(x \to \infty,$ i.e. $n \to \infty,$ and $n > 2m+1) \geq \frac{(n-m)^{m+1}}{(m+1)!}a^{m+1} \geq \frac{n^{m+1}}{2^{m+1}(m+1)!}a^{m+1},$ then $\frac{(1+a)^x}{x^A} \geq \frac{n^{m+1}a^{m+1}}{2^{m+1}(m+1)!(n+1)^m} \geq \frac{na^{m+1}}{2^{2m+1}(m+1)!},$ then $\lim_{x\to\infty}\frac{x^A}{(1+a)^x} = 0,$ *i.e.* $x^A = o((1+a)^x),\ x \to \infty.$ Since $\lim_{x\to\infty}\frac{x^A}{(1+a)^{\epsilon x}} = \lim_{x\to\infty}(\frac{x^{\frac{A}{\epsilon}}}{(1+a)^x})^{\epsilon} = 0,$ then $x^A = o((1+a)^{\epsilon x}),\ x \to \infty.$

Problem. /Tech tool/
Let $a_n = O(b_n),\ n \geq 1,$ and series $\sum_{n=1}^{\infty} b_n < \infty,$ then $\exists C,$ s.t. $\sum_{k=1}^{n} a_k = C + O\left(\sum_{k=n+1}^{\infty} b_k\right).$

> *Solution.* Obviously, series $\sum_{k=1}^{\infty} a_k$ converges, then $\exists C$ s.t. $\sum_{k=1}^{\infty} a_k = C,$ then $\sum_{k=1}^{n} a_k = \sum_{k=1}^{\infty} a_k - \sum_{k=n+1}^{\infty} a_k = C + O\left(\sum_{k=n+1}^{\infty} b_k\right).$